

General Relativity HW-3

PROBLEM 1. Prove by a direct writing of transformations of connections that a Riemann tensor is indeed a tensor.

SOLUTION:

Notation: denote the Jacobian $\frac{\partial x^i}{\partial q^{i'}}$ as $\partial_{i'}^i, \frac{\partial^2 x^i}{\partial q^{j'} \partial q^{k'}}$ as $(\partial^2)_{j'k'}^i$ for simplicity. All indices that appear in the q coordinate will be with $'$ in order to avoid misunderstanding.

We do not assume here that the connection is Levi-Civita connection.

We know that Riemann tensor is defined

$$R^l_{ijk} V^i = [\mathcal{D}_j, \mathcal{D}_k] V^l$$

and by some tedious calculation we know that

$$R^l_{ijk} = \partial_j \Gamma^l_{ik} - \partial_i \Gamma^l_{jk} + \Gamma^m_{ik} \Gamma^l_{jm} - \Gamma^m_{jk} \Gamma^l_{im}$$

Now suppose there is a coordinate transformation $x^i \rightarrow q^{i'}$, and we already know by the covariance of $\mathcal{D}$, that:

$$\Gamma^{i'}_{j'k'} = \partial_{j'}^j \partial_{k'}^k \partial_i^{i'} \Gamma^i_{jk} + (\partial^2)_{j'k'}^i \partial_i^{i'}$$

Then we are ready to derive the tedious coordinate transformation for Riemann tensor.

$$\begin{aligned} R^{l'}_{i'j'k'} &= \partial_{j'} \Gamma^{l'}_{i'k'} - \partial_{i'} \Gamma^{l'}_{j'k'} + \Gamma^{m'}_{i'k'} \Gamma^{l'}_{j'm'} - \Gamma^{m'}_{j'k'} \Gamma^{l'}_{i'm'} \\ &= \partial_{j'}^j \partial_{i'}^i \partial_{k'}^k \partial_l^{l'} \Gamma^l_{ik} + (\partial^2)_{i'k'}^l \partial_l^{l'} - \partial_{i'}^i \partial_{j'}^j \partial_{k'}^k \partial_l^{l'} \Gamma^l_{jk} + (\partial^2)_{j'k'}^l \partial_l^{l'} + \\ &(\partial_{i'}^i \partial_{k'}^k \partial_m^{m'} \Gamma^m_{ik} + (\partial^2)_{i'k'}^m \partial_m^{m'}) (\partial_{j'}^j \partial_{m'}^m \partial_l^{l'} \Gamma^l_{jm} + (\partial^2)_{j'm'}^l \partial_l^{l'}) - (\partial_{j'}^j \partial_{k'}^k \partial_i^{i'} \Gamma^i_{jk} + (\partial^2)_{j'k'}^i \partial_i^{i'}) (\partial_{i'}^i \partial_{m'}^m \partial_l^{l'} \Gamma^l_{im} + (\partial^2)_{i'm'}^l \partial_l^{l'}) \\ &\quad \equiv \partial_{j'}^j \partial_{i'}^i \partial_{k'}^k \partial_l^{l'} (\partial_j \Gamma^l_{ik} - \partial_i \Gamma^l_{jk} + \Gamma^m_{ik} \Gamma^j_{jm} - \Gamma^m_{jk} \Gamma^j_{im}) \\ &\quad + ((\partial^2)_{i'j'}^i \partial_{k'}^k \partial_l^{l'} + \partial_{i'}^i (\partial^2)_{j'k'}^k \partial_l^{l'} + \partial_{i'}^i \partial_{k'}^k (\partial^2)_{lj'}^l) \Gamma^l_{ik} + (\partial^3)_{i'j'k'}^l \partial_l^{l'} + (\partial^2)_{i'k'}^l (\partial^2)_{lj'}^l \\ &\quad - \left[((\partial^2)_{j'i'}^j \partial_{k'}^k \partial_l^{l'} + \partial_{j'}^j (\partial^2)_{i'k'}^k \partial_l^{l'} + \partial_{j'}^j \partial_{k'}^k (\partial^2)_{li'}^l) \Gamma^l_{jk} + (\partial^3)_{i'j'k'}^l \partial_l^{l'} + (\partial^2)_{j'k'}^l (\partial^2)_{li'}^l \right] \\ &\quad + (\partial^2)_{i'k'}^m \partial_{j'}^j \partial_l^{l'} \Gamma^l_{jm} + (\partial^2)_{j'm'}^l \partial_l^{l'} \partial_{i'}^i \partial_{k'}^k \partial_m^{m'} \Gamma^m_{ik} + (\partial^2)_{j'm'}^l \partial_l^{l'} (\partial^2)_{i'k'}^m \partial_m^{m'} \\ &\quad - \left[(\partial^2)_{j'k'}^m \partial_{i'}^i \partial_l^{l'} \Gamma^l_{im} + (\partial^2)_{i'm'}^l \partial_l^{l'} \partial_{j'}^j \partial_{k'}^k \partial_m^{m'} \Gamma^m_{jk} + (\partial^2)_{i'm'}^l \partial_l^{l'} (\partial^2)_{j'k'}^m \partial_m^{m'} \right] \end{aligned}$$

By switching some dummy indices, we obtain a simpler expression:

$$\begin{aligned} &\equiv \partial_{j'}^j \partial_{i'}^i \partial_{k'}^k \partial_l^{l'} R^l_{ijk} \\ &\quad + (2\partial_{i'}^i \partial_{k'}^k (\partial^2)_{lj'}^l + (\partial^2)_{i'j'}^i \partial_{k'}^k \partial_l^{l'}) \Gamma^l_{ik} - (2\partial_{j'}^j \partial_{k'}^k (\partial^2)_{li'}^l + (\partial^2)_{j'i'}^j \partial_{k'}^k \partial_l^{l'}) \Gamma^l_{jk} \\ &\quad + (\partial^2)_{i'm}^l (\partial^2)_{lj'}^l \partial_m^m - (\partial^2)_{i'm}^l (\partial^2)_{j'k'}^m \partial_l^{l'} + (\partial^2)_{j'm}^l \partial_l^{l'} (\partial^2)_{i'k'}^m - (\partial^2)_{j'm}^l (\partial^2)_{li'}^l \partial_k^k \end{aligned}$$

Let's consider the value of 2,3 row of the above result.

The second row: Notice that

$$2\partial_{i'}^i \partial_{k'}^k (\partial^2)_{lj'}^{l'} + (\partial^2)_{i'j'}^i \partial_{k'}^k \partial_l^{l'} = \partial_{k'}^k (\partial_{i'j'} (\partial^i \partial_l^{l'}) - \partial^i (\partial^3)_{i'j'l}^{l'})$$

Thus by switching the index j in the second part of the second row into i gives:

$$\begin{aligned} & (2\partial_{i'}^i \partial_{k'}^k (\partial^2)_{lj'}^{l'} + (\partial^2)_{i'j'}^i \partial_{k'}^k \partial_l^{l'}) \Gamma_{ik}^l - (2\partial_{j'}^j \partial_{k'}^k (\partial^2)_{li'}^{l'} + (\partial^2)_{j'i'}^j \partial_{k'}^k \partial_l^{l'}) \Gamma_{jk}^l \\ &= \left[\partial_{k'}^k (\partial_{i'j'} (\partial^i \partial_l^{l'}) - \partial^i (\partial^3)_{i'j'l}^{l'}) - \partial_{k'}^k (\partial_{j'i'} (\partial^j \partial_l^{l'}) - \partial^j (\partial^3)_{j'i'l}^{l'}) \right] \Gamma_{ik}^l = 0 \end{aligned}$$

which is because that we have put all i' and j' on the commutable position (meaning that switching i', j' doesn't change the value), and it's obvious that the above value is a commutator of i', j' .

The third row: Notice that

$$\begin{aligned} & (\partial^2)_{i'm}^l (\partial^2)_{lj'}^{l'} \partial_{k'}^m - (\partial^2)_{i'm}^l (\partial^2)_{j'k'}^m \partial_l^{l'} = (\partial^2)_{i'm}^l (\partial_{j'} (\partial_l^{l'} \partial_{k'}^m) - 2(\partial^2)_{j'k'}^m \partial_l^{l'}) \\ &= \frac{1}{2} \left[\partial_{i'j'} (\partial_m^l \partial_l^{l'} \partial_{k'}^m) - ((\partial^3)_{i'j'm}^l) \partial_l^{l'} \partial_{k'}^m - \partial_m^l \partial_{i'j'} (\partial_l^{l'} \partial_{k'}^m) \right] - \partial_l^{l'} \left[\partial_{i'j'} (\partial_m^l \partial_{k'}^m) - (\partial^3)_{i'j'm}^l \partial_{k'}^m - \partial_m^l (\partial^3)_{i'j'k'}^m \right] \end{aligned}$$

Thus the third row is:

$$\begin{aligned} & (\partial^2)_{i'm}^l (\partial^2)_{lj'}^{l'} \partial_{k'}^m - (\partial^2)_{i'm}^l (\partial^2)_{j'k'}^m \partial_l^{l'} + (\partial^2)_{j'm}^l \partial_l^{l'} (\partial^2)_{i'k'}^m - (\partial^2)_{j'm}^l (\partial^2)_{li'}^{l'} \partial_{k'}^m \\ &= \frac{1}{2} \left[\partial_{[i'j']} (\partial_m^l \partial_l^{l'} \partial_{k'}^m) - ((\partial^3)_{[i'j']m}^l) \partial_l^{l'} \partial_{k'}^m - \partial_m^l \partial_{[i'j']} (\partial_l^{l'} \partial_{k'}^m) \right] - \partial_l^{l'} \left[\partial_{[i'j']} (\partial_m^l \partial_{k'}^m) - (\partial^3)_{[i'j']m}^l \partial_{k'}^m - \partial_m^l (\partial^3)_{[i'j']k'}^m \right] = 0 \end{aligned}$$

Because all i', j' are on the commutable position.

In conclusion,

$$R_{i'j'k'}^{l'} = \partial_{j'}^j \partial_{i'}^i \partial_{k'}^k \partial_l^{l'} R_{ijk}^l$$

which proves that R is a tensor. □

PROBLEM 2. Prove the property of a cyclic permutation of indexes of Riemann tensor that:

$$R_{\mu[\alpha\nu\beta]} \equiv R_{\mu\alpha\nu\beta} + R_{\mu\beta\alpha\nu} + R_{\mu\nu\beta\alpha} = 0$$

SOLUTION:

By expanding the whole formula gives:

$$\begin{aligned} R_{\mu[ijk]} &= g_{\mu l} [(\partial_j \Gamma_{ik}^l - \partial_i \Gamma_{jk}^l + \Gamma_{ik}^m \Gamma_{jm}^l - \Gamma_{jk}^m \Gamma_{im}^l) + (\partial_k \Gamma_{ji}^l - \partial_j \Gamma_{ki}^l + \Gamma_{ji}^m \Gamma_{km}^l - \Gamma_{ki}^m \Gamma_{jm}^l) + \\ & \quad (\partial_i \Gamma_{kj}^l - \partial_k \Gamma_{ij}^l + \Gamma_{kj}^m \Gamma_{im}^l - \Gamma_{ij}^m \Gamma_{km}^l)] \\ &= g_{\mu l} [\partial_i (\Gamma_{kj}^l - \Gamma_{jk}^l) + \partial_j (\Gamma_{ik}^l - \Gamma_{ki}^l) + \partial_k (\Gamma_{ji}^l - \Gamma_{ij}^l) + \Gamma_{im}^l (\Gamma_{kj}^m - \Gamma_{jk}^m) - \Gamma_{jm}^l (\Gamma_{ik}^m - \Gamma_{ki}^m) + \Gamma_{km}^l (\Gamma_{ji}^m - \Gamma_{ij}^m)] \end{aligned}$$

Because we assume that the connection is torsion-free, while each of the term in the small bracket has an anti-symmetric form with respect to the connection, thus

$$R_{\mu[ijk]} = 0$$

□

PROBLEM 3. Naively Riemann tensor has n^4 independent components in n dimensions since it has four indexes. Compute how many independent indexes it has accounting all its algebraic symmetries ($R_{\mu\nu\alpha\beta} = R_{\alpha\beta\mu\nu} = -R_{\nu\mu\alpha\beta} = -R_{\mu\nu\beta\alpha}$ and the cyclic property from the previous task). Does this result change upon accounting Bianchi identity?

SOLUTION:

For fixed jk , R_{lijk} is anti-symmetric with respect to l, i , thus the independent components considering this is $n^2 \binom{n}{2}$. Considering j, k lowers the numbers of independent components into $\binom{n}{2}^2$.

When considering the symmetric property with respect to l, i and j, k , it is equivalent to think a symmetric rank 2 tensor with dimension $\binom{n}{2}$. Thus by homework 1, the independent components are

$$\frac{(\frac{n^2-n}{2} + 1) \frac{n^2-n}{2}}{2} = \frac{(n^2 - n + 2)(n^2 - n)}{8}$$

When considering the first Bianchi identity

$$R_{\mu[\alpha\nu\beta]} = 0$$

Notice that this constraint only works (meaning does not repeat with previous constraints) when the four indices in the tensor are all different (we can easily verify the situation when the indices are like $accb$, the first Bianchi identity is automatically satisfied. For other situations of existing repeated indices, they are trivial). Thus the reduced DOF is $\binom{n}{4}$, thus the left DOF is:

$$\frac{(n^2 - n + 2)(n^2 - n)}{8} - \frac{n(n-1)(n-2)(n-3)}{24} = \frac{1}{12}n^4 - \frac{1}{12}n^2$$

I think when considering the second Bianchi identity, the result won't change. Because the second Bianchi identity applies for any tensor with the same rank, so there is no extra constraints over the Riemann tensor.